NỀN TẢNG AI TẠO SINH

(IT5410 – Foundation of Generative AI)

MÔ HÌNH SINH SÂU

VAE

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Trường CNTT&TT, ĐHBKHN

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Nội dung

- Mở đầu
- Một số vấn đề của Học sâu
- Một số kiến trúc mạng nơron
- **Mô hình sinh sâu**
- Đánh giá chất lượng
- Học tăng cường

Some successes: Image + Video generation

Generative Models

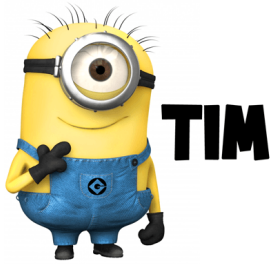
- Probabilistic models of data
- Sample: lấy mẫu dữ liệu (sinh/tạo ra dữ liệu)
- Evaluate likelihood: tính likelihood của dữ liệu cho trước
- Train: huấn luyện
- Representation: biểu diễn mới
- What if all we care about is sampling?
  - *Not in the training data, but the novel samples.*

Probabilistic models

Introduction

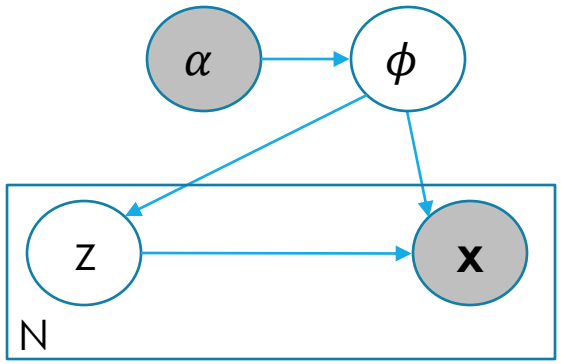
Probabilistic model

- Our assumption on how the data samples were generated
  (giả thuyết của chúng ta về quá trình mà các mẫu dữ liệu đã được sinh ra như thế nào)
- Example: how a sentence is generated?
  - We assume our brain does as follow:
  - *First choose the topic of the sentence*
  - *Generate the words one-by-one to form the sentence*
- How will TIM be drawn?



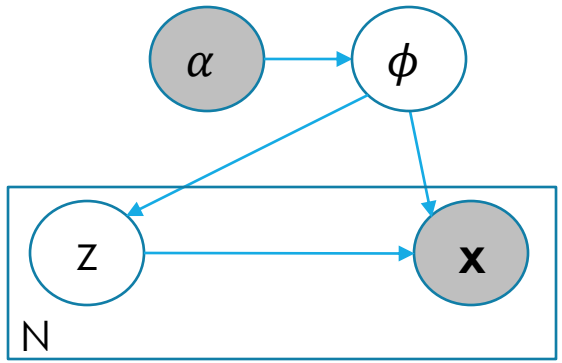
Probabilistic model

- A model sometimes consists of
  - **Observed variable** (e.g., $\boldsymbol{x}$) which models the observation (data instance) (biến quan sát được)
  - **Hidden variable** which describes the hidden things (e.g., z, ϕ) (biến ẩn)
  - **Relations** between the variables
- Each variable follows some probability distribution (mỗi biến tuân theo một phân bố xác suất nào đó)



Different types of models

- Probabilistic graphical model (PGM): Graph + Probability Theory
  (mô hình đồ thị xác suất)
  - Each vertex represents a random variable,
    grey circle means “observed”,
    white circle means “latent”
  - Each edge represents the conditional
    dependence between two variables



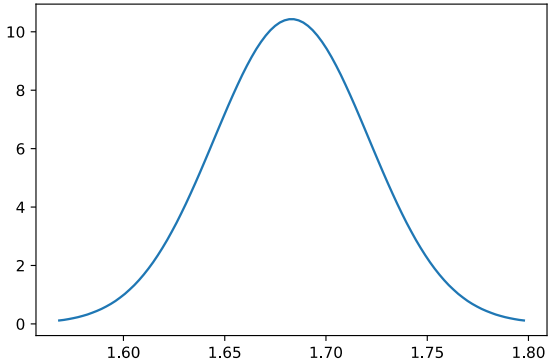
- Latent variable model: a PGM which has at least one latent variable
- **Generative model:** a model that enables us to generate data instances

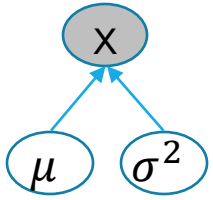
Univariate normal distribution

- We wish to know the average height of a person
  - We had collected a dataset from 10 people in Hanoi:
    D = {1.6, 1.7, 1.65, 1.63, 1.75, 1.71, 1.68, 1.72, 1.77, 1.62}
- Let x denote the random variable that represents the height of a person
- **Assumption:** x follows a Normal distribution (Gaussian) with the following *probability density function* (PDF)

$$\mathcal{N}(x|\mu,\sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}$$

  - where $\{\mu,\sigma^2\}$ are the mean and variance
- Note:
  - $\mathcal{N}(x|\mu,\sigma^2)$ represents the class of normal distributions
  - This class is parameterized by $\boldsymbol{\theta} = (\mu,\sigma^2)$
- **Learning:** we need to know specific values of $\{\mu,\sigma^2\}$



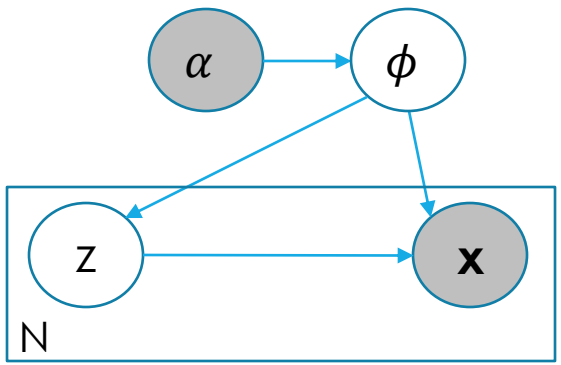


PGM: some well-known models

- Gaussian mixture model (GMM)
  - Modeling real-valued data
- Latent Dirichlet allocation (LDA)
  - Modeling the topics hidden in textual data
- Hidden Markov model (HMM)
  - Modeling time-series, i.e., data with time stamps or sequential nature
- Conditional Random Field (CRF)
  - for structured prediction
- Deep generative models
  - Modeling the hidden structures, generating artificial data

Probabilistic model: inference & learning

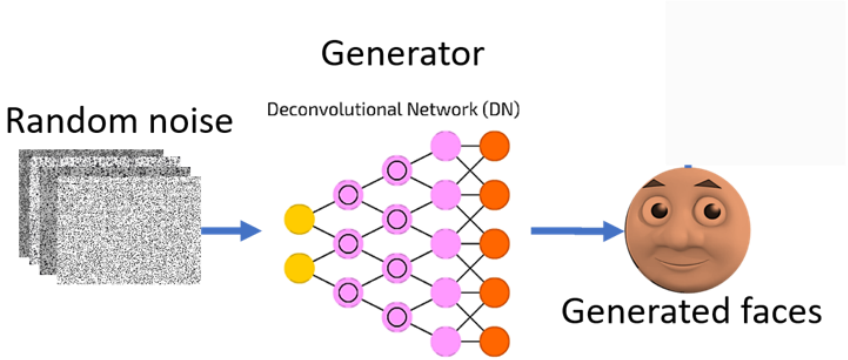
- **Inference** for a given instance $\boldsymbol{x}_n$
  (Suy diễn/phán đoán đối với một quan sát cho trước)
  - Recovery of the local variable (e.g., z_n), or
  - The distribution of the local variables (e.g., $P(z_n|\phi, \boldsymbol{x}_n)$)
  - Example: for GMM, we want to know z_n indicating which Gaussian did generate $\boldsymbol{x}_n$



- **Learning (estimation)**
  (Học/ước lượng mô hình)
  - Given a training dataset, estimate the joint distribution of the variables
    - E.g., estimate the density function $p(\phi, z_1, \dots, z_n, \boldsymbol{x}_1, \dots, \boldsymbol{x}_n|\alpha)$
    - E.g., estimate $P(\boldsymbol{x}_1, \dots, \boldsymbol{x}_n|\alpha)$
    - E.g., estimate α
    - Inference of local variables is often needed

Generative model: sampling

- **Sampling** data
  - Make novel data samples, given a trained model
    (tạo ra dữ liệu mới từ mô hình đã có)



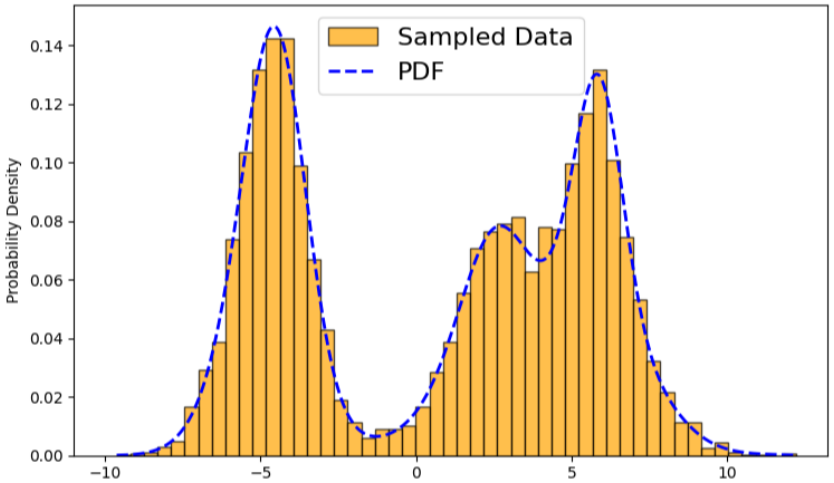
- Application:
  - Entertainment (ngành giải trí): videos, images, musics, …
  - Limited resources: khi khả năng thu thập được ít mẫu dữ liệu
  - Fashion: tạo mẫu quần/áo thời trang
  - Design: tạo mẫu trang thiết bị mới
  - Materials: tạo các vật liệu mới
  - …

Generative models

Learning

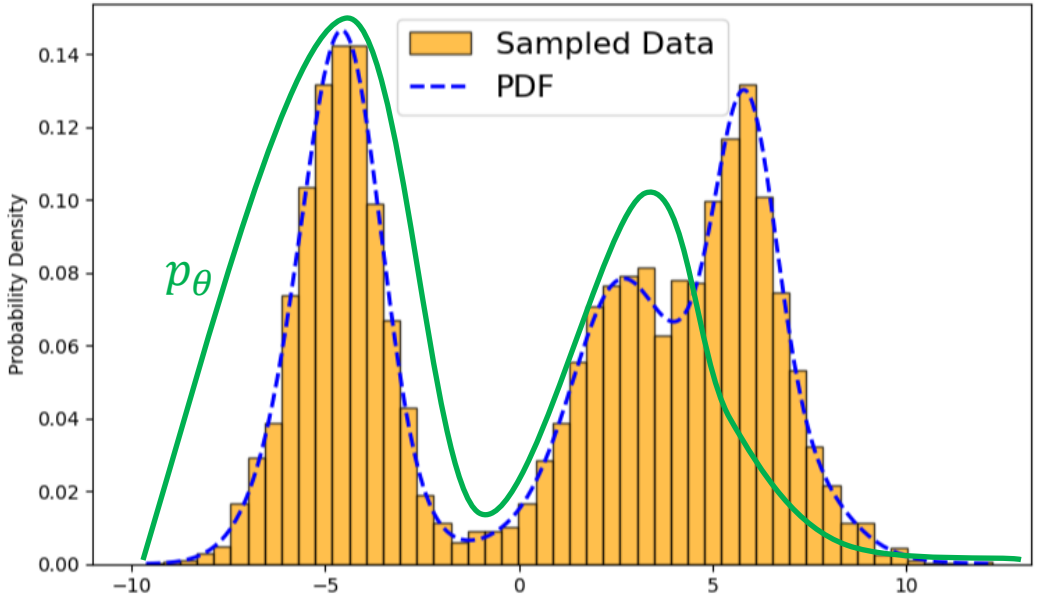
Learning a generative model

- Given a training set of examples, e.g., images of dogs
- We want to learn a probability distribution P(**x**) over images **x** such that
  - **Generation:** If we sample $\boldsymbol{x}_{new} \sim P(\boldsymbol{x})$, $\boldsymbol{x}_{new}$ should look like a dog (*sampling*)
  - **Density estimation:** P(**x**)
  - **Unsupervised representation learning:** We should be able to learn what these images have in common, e.g., ears, tail, etc. (*features*)



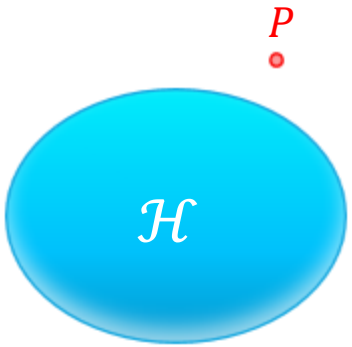
The sampling distribution

- Dataset **D** = {$\mathbf{x}_1$, $\mathbf{x}_2$, …, $\mathbf{x}_m$}
- **Hardness** of the learning problem:
  - P(**x**) in the space of all probability distributions
- In practice, we often find a $P_\theta(\boldsymbol{x})$ to **approximate** $P(\boldsymbol{x})$



Hypothesis space

- Ussually, we can choose a restricted set $\mathcal{H}$ of distributions
  - Parameterized by $\theta \in \Theta$
  - A learner must find one $P_\theta \in \mathcal{H}$



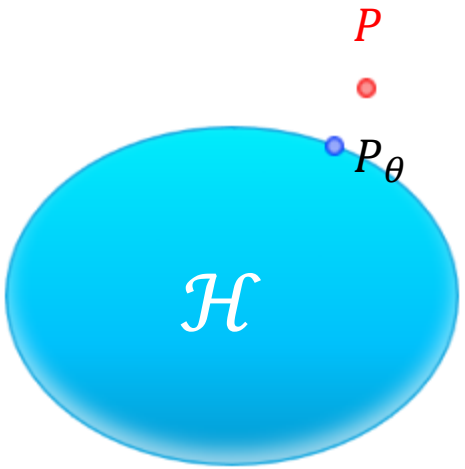
- Hypothesis space *(model family)*:

  a set $\mathcal{H}$ of distributions, providing candidates for a learner
  - Represents prior knowledge about a task
  - Represents our ***inductive bias*** or *preference*
- Each P_θ is often called a "**model**"
- Gaussian family:

$$\mathcal{H} = \{P_\theta : P_\theta \text{ is the normal distribution with } \theta = (\mu, \sigma), \mu \in \mathbb{R}, \sigma \in \mathbb{R}_+\}$$

Learning goal

- Find a model P_θ that precisely captures the distribution P from which our data was sampled
- Intractability:
  - P(**x**) is in the space of all probability distributions
  - The sampled data set is limited
  - Computational reasons
- We want to select P_θ to be the ”best” approximation to the underlying distribution P
  - **What is “best”?**
  - Depends on specific task of interest



Learning as density estimation

- We want to learn the full distribution so that later we can answer any probabilistic inference query
- In this setting we can view the learning problem as **density estimation**
- We want to construct P_θ as **"close"** as possible to P
  (recall we assume we are given a dataset **D** of samples from P)

How do we evaluate "closeness"?

KL-divergence

- How should we measure distance between distributions?
- The **Kullback-Leibler divergence** (KL-divergence) between two distributions P and Q is defined as

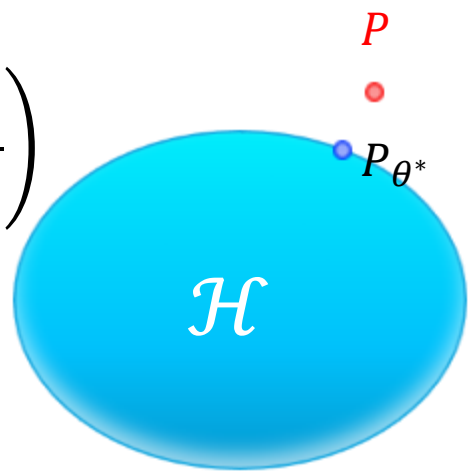
$$KL(P||Q) = \mathbb{E}_{\boldsymbol{x}\sim P(\boldsymbol{x})}\left(\log\frac{p(\boldsymbol{x})}{q(\boldsymbol{x})}\right)$$

  - where $\mathrm{p}(\boldsymbol{x})$ and $\mathrm{q}(\boldsymbol{x})$ represents the densities of P and Q, respectively
- Note that:
  - $KL(P||Q) \geq 0$ for any P and Q, and $KL(P||P) = 0$
  - $KL(P||Q) \neq KL(Q||P)$
- It measures the loss (in bits) when describing distribution P by Q.

Learning: a revisit

- We want to construct P_θ as **"close"** as possible to P
  (Given a dataset **D** of samples from P)
- Closeness by KL:

$$KL(P||P_\theta) = \mathbb{E}_{\boldsymbol{x}\sim P(\boldsymbol{x})}\left(\log\frac{p(\boldsymbol{x})}{p_\theta(\boldsymbol{x})}\right)$$



- Learning by minimizing $KL(P||P_\theta)$

$$\theta^* = \underset{\theta\in\Theta}{\operatorname{argmin}}\, KL(P||P_\theta)$$

  - Find the parameter θ^* that minimizes $KL(P||P_\theta)$
  - θ^* provides the minimal loss when compressing P by P_{θ^*}

Expected log-likelihood

- We can rewrite

$$KL(P||P_\theta) = \mathbb{E}_{\boldsymbol{x}\sim P(\boldsymbol{x})}\left(\log\frac{p(\boldsymbol{x})}{p_\theta(\boldsymbol{x})}\right) = \mathbb{E}_{\boldsymbol{x}\sim P(\boldsymbol{x})}(\log p(\boldsymbol{x})) - \mathbb{E}_{\boldsymbol{x}\sim P(\boldsymbol{x})}(\log p_\theta(\boldsymbol{x}))$$

  - The first term does not depend on θ
- Minimizing KL is equivalent to maximizing the *Expected log-likelihood* $\mathbb{E}_{\boldsymbol{x}}(\log p_\theta(\boldsymbol{x}))$
- Learning can be done by **Maximum Likelihood Estimation (MLE)**

$$\theta^* = \underset{\theta\in\Theta}{\operatorname{argmax}}\, \mathbb{E}_{\boldsymbol{x}\sim P(\boldsymbol{x})}(\log p_\theta(\boldsymbol{x}))$$

  - In general, we do not know P
  - So, we cannot access to the objective

Maximum likelihood

- We approximate the *expected log-likelihood* $\mathbb{E}_{\boldsymbol{x}\sim P(\boldsymbol{x})}(\log p_\theta(\boldsymbol{x}))$ by

$$\mathbb{E}_{\boldsymbol{x}\in\boldsymbol{D}}(\log p_\theta(\boldsymbol{x})) = \frac{1}{m}\sum_{\boldsymbol{x}\in\boldsymbol{D}} \log p_\theta(\boldsymbol{x})$$

  - Sometimes known as *Empirical log-likelihood* (note the similarity with empirical loss in ML)

- MLE is the formulated as

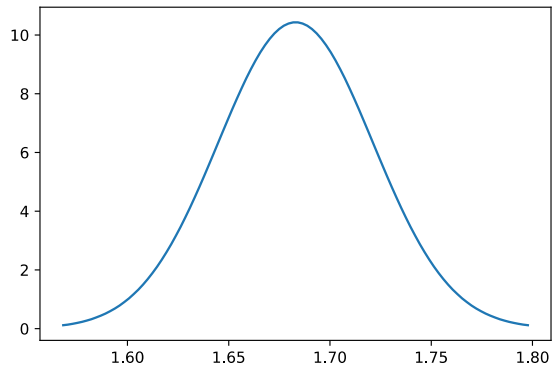
$$\theta^* = \underset{\theta\in\Theta}{\operatorname{argmax}} \frac{1}{m}\sum_{\boldsymbol{x}\in\boldsymbol{D}} \log p_\theta(\boldsymbol{x})$$

  - This is equivalent to maximizing the likelihood $P(\boldsymbol{x}_1, \dots, \boldsymbol{x}_m) = \prod_{i=1}^{m} P(\boldsymbol{x}_i)$ for i.i.d. samples

MLE: Gaussian example (1)

- We wish to estimate the height of a person in the world.
- Use a dataset **D** = {1.6, 1.7, 1.65, 1.63, 1.75, 1.71, 1.68, 1.72, 1.77, 1.62}
  - Let x be the random variable representing the height of a person.
  - Model: assume that x follows a Gaussian distribution with ***unknown*** mean μ and variance σ^2
  - **Learning:** estimate (μ, σ) from the given data $\boldsymbol{D} = \{x_1, \dots, x_{10}\}$.
- Let $f(x|\mu, \sigma)$ be the density function of the Gaussian family, parameterized by (μ, σ).
  - $f(x_n|\mu, \sigma)$ is the likelihood of instance x_n.
  - $f(\boldsymbol{D}|\mu, \sigma)$ is the likelihood function of **D**.
- Using MLE, we will find

$$(\mu_*, \sigma_*) = \arg\max_{\mu,\sigma} f(\boldsymbol{D}|\mu, \sigma)$$



MLE: Gaussian example (2)

- **i.i.d assumption:** we assume that the data are independent and identically distributed (dữ liệu được sinh ra một cách độc lập)
  - As a result, we have $P(\boldsymbol{D}|\mu,\sigma) = P(x_1,\dots,x_{10}|\mu,\sigma) = \prod_{i=1}^{10} P(x_i|\mu,\sigma)$
- Using this assumption, MLE will be

$$(\mu_*,\sigma_*) = \arg\max_{\mu,\sigma} \prod_{i=1}^{10} f(x_i|\mu,\sigma) = \arg\max_{\mu,\sigma} \prod_{i=1}^{10} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x_i-\mu)^2}$$

$$= \arg\max_{\mu,\sigma} \log \prod_{i=1}^{10} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x_i-\mu)^2}$$

Log trick, $\log \overset{\text{def}}{=} \ln$

$$= \arg\max_{\mu,\sigma} \sum_{i=1}^{10} \left(-\frac{1}{2\sigma^2}(x_i-\mu)^2 - \log\sqrt{2\pi\sigma^2} \right)$$

- Using gradients (w.r.t μ, σ), we can find

$$\mu_* = \frac{1}{10}\sum_{i=1}^{10} x_i = 1.683, \qquad \sigma_*^2 = \frac{1}{10}\sum_{i=1}^{10}(x_i-\mu_*)^2 \approx 0.0015$$

Generative models

Approximation by mixture models

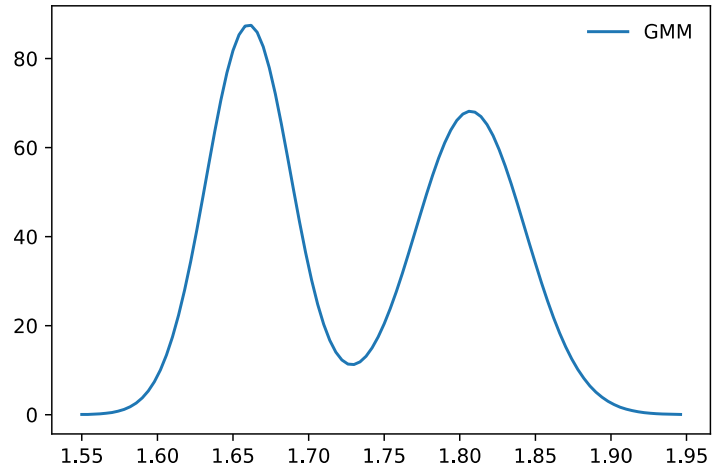
Learning the data distribution

- Dataset **D** = {$\mathbf{x}_1$, $\mathbf{x}_2$, …, $\mathbf{x}_m$}
  - Images about dogs
- **Hardness** of the learning problem:
  - P(**x**) is in the space of all probability distributions
- In practice
  - We often find a $P_\theta(\boldsymbol{x})$ to **approximate** $P(\boldsymbol{x})$
- How to choose a good model family?
  - Gaussian family? => too simple

Gaussian mixture model (GMM)

- GMM: we assume that the data are samples from *K* Gaussian distributions.
  - Each instance **x** is generated from one of those K Gaussians by the following ***generative process***:
    - *Take the component index* $z \sim Categorical(\boldsymbol{\phi})$
    - *Generate* $\boldsymbol{x} \sim Normal(\boldsymbol{\mu}_z, \boldsymbol{\Sigma}_z)$
- The density function is

$$q(\boldsymbol{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}, \boldsymbol{\phi}) = \sum_{k=1}^{K} \phi_k \mathcal{N}(\boldsymbol{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$$

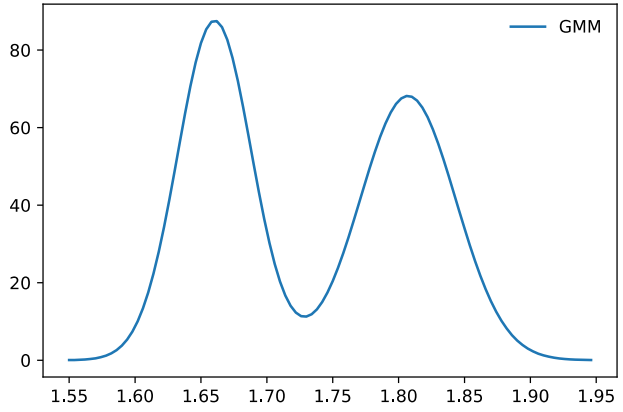


  - $\boldsymbol{\phi} = (\phi_1, \ldots, \phi_K)$ represents the weights of the Gaussians: $\sum_{k=1}^{K} \phi_k = 1, \quad \phi_j \geq 0, \ \forall j$
  - Each Gaussian has density $\mathcal{N}(\boldsymbol{x} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{\sqrt{\det(2\pi\boldsymbol{\Sigma})}} \exp\left[-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})\right]$

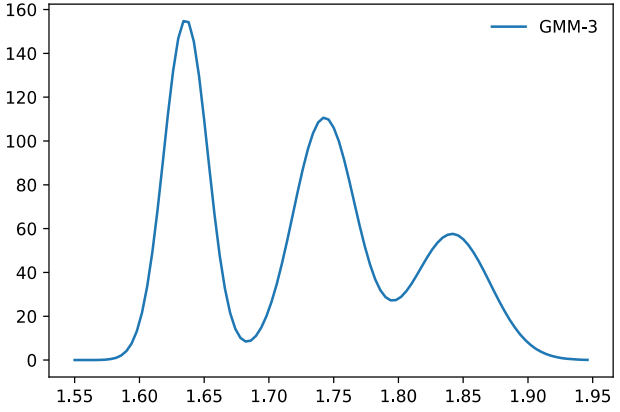
- Note: z is an unobserved (latent) variable, x is observable

GMM: approximation ability

- The density $q(\boldsymbol{x}|\boldsymbol{\mu},\boldsymbol{\Sigma},\boldsymbol{\phi}) = \sum_{k=1}^{K} \phi_k \mathcal{N}(\boldsymbol{x} \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$
  - Gaussian model: $K = 1$ component
- A larger K produces a more complex model Q



GMM with 2 components



GMM with 3 components

- **GMMs are universal approximators**
  - Any smooth density can be approximated arbitrarily well by a GMM with enough components

Dalal, S. R., and W. J. Hall. "Approximating Priors by Mixtures of Natural Conjugate Priors." J. of the Royal Statistical Society. Series B (Methodological), vol. 45, no. 2, 1983, pp. 278–286.

Infinite GMM

- *Mixture of an infinite number of Gaussians:* we assume that the data are samples from an infinite number of Gaussians
  - Each instance **x** is generated from one of those Gaussians by the following ***generative process***:
    - *Choose* $\boldsymbol{z} \sim Normal(0, \boldsymbol{I})$ $\quad P(\mathbf{z})$
    - *Generate* $\boldsymbol{x} \sim Normal(\boldsymbol{\mu}_\theta(\boldsymbol{z}), \boldsymbol{\Sigma}_\theta(\boldsymbol{z}))$ $\quad P(\mathbf{x} \mid \mathbf{z})$
      - Where $\boldsymbol{\mu}_\theta, \boldsymbol{\Sigma}_\theta$ are neural networks, parameterized by θ
- **Universal approximator?**
- Each component is simple, but the marginal P(**x**) is very complex

Variational auto-encoder

Variational inference,
Amortized inference,
Sampling

Learning for GMM

- Learning by MLE:

$$\theta^* = \underset{\theta}{\operatorname{argmax}} \frac{1}{m} \sum_{\boldsymbol{x} \in \boldsymbol{D}} \log p_\theta(\boldsymbol{x})$$

  - where $p_\theta(\boldsymbol{x}) = \sum_{k=1}^{K} \phi_k \frac{1}{\sqrt{\det(2\pi \boldsymbol{\Sigma}_k)}} \exp\left[-\frac{1}{2}(\boldsymbol{x} - \boldsymbol{\mu}_k)^T \boldsymbol{\Sigma}_k^{-1} (\boldsymbol{x} - \boldsymbol{\mu}_k)\right]$, $\theta = (\boldsymbol{\phi}, \boldsymbol{\mu}, \boldsymbol{\Sigma})$

- Evaluation of $\log p_\theta(\boldsymbol{x})$ is **hard** in general, since

$$\log p_\theta(\boldsymbol{x}) = \log \sum_{\text{All possible values of } \boldsymbol{z}} p_\theta(\boldsymbol{x}, \boldsymbol{z})$$

  - E.g., for $\boldsymbol{z} \in \{0,1\}^{100}$, the sum has 2^{100} terms

- It is even harder for more complex models

➔ Approximation is needed

Evidence Lower Bound

- Note

$$\log p_\theta(\boldsymbol{x}) = \log \sum_{\boldsymbol{z}\in\mathcal{Z}} p_\theta(\boldsymbol{x},\boldsymbol{z}) = \log \sum_{\boldsymbol{z}\in\mathcal{Z}} \frac{q(\boldsymbol{z})}{q(\boldsymbol{z})} p_\theta(\boldsymbol{x},\boldsymbol{z}) = \log \mathbb{E}_{q(\boldsymbol{z})} \frac{p_\theta(\boldsymbol{x},\boldsymbol{z})}{q(\boldsymbol{z})}$$

- Since log is concave, Jensen Inequality suggests

$$\log \mathbb{E}_{q(\boldsymbol{z})} \frac{p_\theta(\boldsymbol{x},\boldsymbol{z})}{q(\boldsymbol{z})} \geq \mathbb{E}_{q(\boldsymbol{z})} \log \frac{p_\theta(\boldsymbol{x},\boldsymbol{z})}{q(\boldsymbol{z})} = \mathbb{E}_{q(\boldsymbol{z})} \log p_\theta(\boldsymbol{x},\boldsymbol{z}) - \mathbb{E}_{q(\boldsymbol{z})} \log q(\boldsymbol{z})$$

- This is called the Evidence Lower Bound (**ELBO**)
- For any $q(\boldsymbol{z})$:

$$\log p_\theta(\boldsymbol{x}) \geq ELBO$$

  - For $\mathrm{ELBO} = \mathbb{E}_{q(\boldsymbol{z})} \log p_\theta(\boldsymbol{x},\boldsymbol{z}) - \mathbb{E}_{q(\boldsymbol{z})} \log q(\boldsymbol{z})$
- When $q(\boldsymbol{z}) = p_\theta(\boldsymbol{z}|\boldsymbol{x})$:

$$\log p_\theta(\boldsymbol{x}) = \mathbb{E}_{p_\theta(\boldsymbol{z}|\boldsymbol{x})} \log p_\theta(\boldsymbol{x},\boldsymbol{z}) - \mathbb{E}_{p_\theta(\boldsymbol{z}|\boldsymbol{x})} \log p_\theta(\boldsymbol{z}|\boldsymbol{x}) = \boldsymbol{ELBO}$$

Variational inference

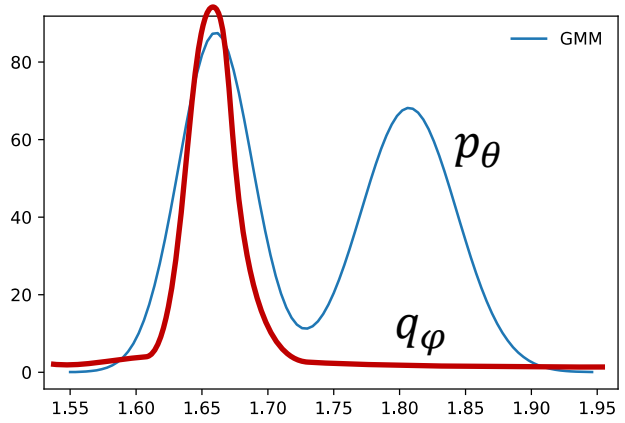
- When the posterior $p_\theta(\boldsymbol{z}|\boldsymbol{x})$ is easy to compute, we can learn the model by maximizing

$$\frac{1}{m}\sum_{\boldsymbol{x}\in\boldsymbol{D}} \log p_\theta(\boldsymbol{x}) = \frac{1}{m}\sum_{\boldsymbol{x}\in\boldsymbol{D}} \left[\mathbb{E}_{p_\theta(\boldsymbol{z}|\boldsymbol{x})} \log p_\theta(\boldsymbol{x},\boldsymbol{z}) - \mathbb{E}_{p_\theta(\boldsymbol{z}|\boldsymbol{x})} \log p_\theta(\boldsymbol{z}|\boldsymbol{x})\right]$$

  - E.g., for the case of GMM

- What if the posterior $p_\theta(\boldsymbol{z}|\boldsymbol{x})$ is intractable to compute?

- **Variational inference (VI):**
  - choose a family of *simple* distributions $q_\varphi(\boldsymbol{z})$, parameterized by φ (variational parameters)
  - then find φ^* so that $q_{\varphi^*}(\boldsymbol{z})$ is as close as possible to $p_\theta(\boldsymbol{z}|\boldsymbol{x})$



VI and KL

- Maximize the ELBO

$$\frac{1}{m}\sum_{i=1}^{m}\left[\mathbb{E}_{q_{\varphi_i}(\boldsymbol{z})}\log p_\theta(\boldsymbol{x}_i,\boldsymbol{z}) - \mathbb{E}_{q_{\varphi_i}(\boldsymbol{z})}\log q_{\varphi_i}(\boldsymbol{z})\right]$$

  - given a training set **D** = {$\mathbf{x}_1$, $\mathbf{x}_2$, …, $\mathbf{x}_m$}

- Maximizing ELBO is equivalent to Minimizing KL, due to

$$\log p_\theta(\boldsymbol{x}) = ELBO + KL(q_\varphi(\boldsymbol{z})||p_\theta(\boldsymbol{z}|\boldsymbol{x}))$$

- Jointly optimize over
  - $\varphi_1, \ldots, \varphi_m$ (variational parameters)
  - θ (model parameters)

VI: some properties

- **Variational inference (VI):**
  - choose a family of *simple* distributions $q_\varphi(\boldsymbol{z})$, parameterized by φ
  - find φ^* so that $q_{\varphi^*}(\boldsymbol{z})$ is as close as possible to $p_\theta(\boldsymbol{z}|\boldsymbol{x})$

- Pros:
  - Easy to be used in a large class of models
  - Efficient in practice
- Cons:
  - Hard to choose a good variational family
    - When we do not know the explicit form for the posterior $p_\theta(\boldsymbol{z}|\boldsymbol{x})$
  - For *inference*, given model param θ and instance $\boldsymbol{x}$, we estimate the posterior $p_\theta(\boldsymbol{z}|\boldsymbol{x})$ *by solving an optimization problem:*

$$\max_{\varphi} \mathbb{E}_{q_\varphi(\boldsymbol{z})} \log p_\theta(\boldsymbol{x}, \boldsymbol{z}) - \mathbb{E}_{q_\varphi(\boldsymbol{z})} \log q_\varphi(\boldsymbol{z})$$

Expensive

  - Require too many variational parameters
    - Each instance $\mathbf{x}_i$ requires one specific φ_i ➔ O(m) parameters
    - GMM needs $O(mKn^2)$ params, where K is #components, n is #dims

Amortized inference

$$\max_{\theta} \frac{1}{m} \sum_{\boldsymbol{x} \in \boldsymbol{D}} \log p_\theta(\boldsymbol{x}) \quad \geq \quad \max_{\theta, \varphi_1, \ldots, \varphi_m} \frac{1}{m} \sum_{\boldsymbol{x}_i \in \boldsymbol{D}} \mathrm{L}(\boldsymbol{x}_i; \theta, \varphi)$$

- Where $\mathrm{L}(\boldsymbol{x}_i; \theta, \varphi) = \mathbb{E}_{q_{\varphi_i}(\boldsymbol{z})} \log p_\theta(\boldsymbol{x}_i, \boldsymbol{z}) - \mathbb{E}_{q_{\varphi_i}(\boldsymbol{z})} \log q_{\varphi_i}(\boldsymbol{z})$

- VI uses φ_i for each point $\mathbf{x}_i$.
  - May not scale well with large datasets; prone to overfitting
- **Amortization:** *we learn a **single** neural network $f_w : \boldsymbol{x} \longmapsto \varphi$ that maps each input **x** to a set of (good) variational parameters*
  - f_w has a *trainable* parameter w
  - For a given input $\mathbf{x}_i$, f_w will produce the parameter $\varphi_i = f_w(\boldsymbol{x}_i)$ of the variational distribution $q_{\varphi_i}(\boldsymbol{z})$
- *Amortized inference:* feed instance **x** to the trained network to get the variational parameter $\varphi = f_w(\boldsymbol{x})$
  - No optimization ➔ cheap

Kingma, D. P. & Welling, M. (2014). Auto-Encoding Variational Bayes. *ICLR*.

Learning with amortized inference

- We can use using stochastic gradient descent to solve

$$\max_{\theta,\varphi_1,\ldots,\varphi_m} \sum_{\boldsymbol{x}_i \in \boldsymbol{D}} \mathrm{L}(\boldsymbol{x}_i;\theta,\varphi)$$

- *Initialize* $\theta^{(0)}, \varphi^{(0)}$
- *At iteration* $j \geq 1$*:*
  - *Randomly sample a data point* $\boldsymbol{x}_i$ *from* $\boldsymbol{D}$
  - *Compute* $\nabla_\theta L\left(\boldsymbol{x}_i;\theta^{(j-1)},\varphi^{(j-1)}\right)$ *and* $\nabla_\varphi L\left(\boldsymbol{x}_i;\theta^{(j-1)},\varphi^{(j-1)}\right)$
  - *Update* $\theta^{(j)}, \varphi^{(j)}$ *in the gradient direction*

- How to compute the gradients?
  - $\mathrm{L}(\boldsymbol{x}_i;\theta,\varphi) = \mathbb{E}_{q_{\varphi_i}(\boldsymbol{z})} \log p_\theta(\boldsymbol{x}_i,\boldsymbol{z}) - \mathbb{E}_{q_{\varphi_i}(\boldsymbol{z})} \log q_{\varphi_i}(\boldsymbol{z})$
  - The expectation complicates gradient computation for φ

Reparameterization trick (mẹo đổi biến)

- Consider **z** being **continuous**, and we want to compute a gradient with respect to φ of

$$\mathbb{E}_{q_\varphi(\boldsymbol{z})}[r(\boldsymbol{z})] = \int q_\varphi(\boldsymbol{z}) r(\boldsymbol{z}) d\boldsymbol{z}$$

- Suppose $q_\varphi(\boldsymbol{z}) = \mathcal{N}(\boldsymbol{\mu}, \sigma^2 \boldsymbol{I})$ is Gaussian with parameters $\varphi = (\boldsymbol{\mu}, \sigma)$
  - Since $\boldsymbol{z} \sim q_\varphi(\boldsymbol{z})$, there exists representation $\boldsymbol{z} = \boldsymbol{\mu} + \sigma\boldsymbol{\epsilon}$ where $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \boldsymbol{I})$
- We can write

$$\mathbb{E}_{\boldsymbol{z} \sim q_\varphi(\boldsymbol{z})}[r(\boldsymbol{z})] = \mathbb{E}_{\boldsymbol{\epsilon} \sim \mathcal{N}(0, \boldsymbol{I})}[r(\boldsymbol{\mu} + \sigma\boldsymbol{\epsilon})]$$

$$\nabla_\varphi \mathbb{E}_{q_\varphi(\boldsymbol{z})}[r(\boldsymbol{z})] = \nabla_\varphi \mathbb{E}_{\boldsymbol{\epsilon}}[r(\boldsymbol{\mu} + \sigma\boldsymbol{\epsilon})] = \mathbb{E}_{\boldsymbol{\epsilon}}\left[\nabla_\varphi r(\boldsymbol{\mu} + \sigma\boldsymbol{\epsilon})\right]$$

- Easy to estimate via Monte Carlo if r is differentiable w.r.t. φ, since $\boldsymbol{\epsilon}$ is easy to sample
  - $\mathbb{E}_{\boldsymbol{\epsilon}}\left[\nabla_\varphi r(\boldsymbol{\mu} + \sigma\boldsymbol{\epsilon})\right] \approx \frac{1}{K}\sum_{j=1}^{K} \nabla_\varphi r\left(\boldsymbol{\mu} + \sigma\boldsymbol{\epsilon}_j\right)$, where $\boldsymbol{\epsilon}_1, \dots, \boldsymbol{\epsilon}_K \sim \mathcal{N}(0, \boldsymbol{I})$

(dùng Monte Carlo để xấp xỉ, nếu hàm r có đạo hàm, vì ta dễ lấy mẫu $\boldsymbol{\epsilon}$)

Variational auto-encoder (VAE)

- Since $q_\varphi(\boldsymbol{z})$ approximates the posterior $p_\theta(\boldsymbol{z}|\boldsymbol{x})$, we can write it as $q_\varphi(\boldsymbol{z}|\boldsymbol{x})$ and

$$\begin{aligned}
\mathrm{L}(\boldsymbol{x};\theta,\varphi) &= \mathbb{E}_{q_\varphi(\boldsymbol{z}|\boldsymbol{x})}\log p_\theta(\boldsymbol{x},\boldsymbol{z}) - \mathbb{E}_{q_\varphi(\boldsymbol{z}|\boldsymbol{x})}\log q_\varphi(\boldsymbol{z}|\boldsymbol{x}) \\
&= \mathbb{E}_{q_\varphi(\boldsymbol{z}|\boldsymbol{x})}\left[\log p_\theta(\boldsymbol{x},\boldsymbol{z}) - \log p_\theta(\boldsymbol{z}) + \log p_\theta(\boldsymbol{z}) - \log q_\varphi(\boldsymbol{z}|\boldsymbol{x})\right] \\
&= \mathbb{E}_{q_\varphi(\boldsymbol{z}|\boldsymbol{x})}[\log p_\theta(\boldsymbol{x}|\boldsymbol{z})] - KL(q_\varphi(\boldsymbol{z}|\boldsymbol{x})\,||p_\theta(\boldsymbol{z}))
\end{aligned}$$

- Maximize L: maximize $p_\theta(\boldsymbol{x}|\boldsymbol{z})$ and push $q_\varphi(\boldsymbol{z}|\boldsymbol{x})$ close to $p_\theta(\boldsymbol{z})$
- **Encoder (bộ mã hoá):**
  - Maps each data point $\boldsymbol{x}$ to a latent vector $\hat{\boldsymbol{z}}$, a sample from a Gaussian ($q_\varphi(\boldsymbol{z}|\boldsymbol{x})$) with parameter $(\mu,\sigma) = Encoder_\varphi(\boldsymbol{x})$
- **Decoder (bộ giải mã):**
  - Reconstruct $\hat{\boldsymbol{x}}$ from a latent vector $\hat{\boldsymbol{z}}$, i.e., pick a sample from a Gaussian ($p_\theta(\boldsymbol{x}|\hat{\boldsymbol{z}})$) with parameter $Decoder_\theta(\hat{\boldsymbol{z}})$

Kingma, D. P. & Welling, M. (2014). Auto-Encoding Variational Bayes. *ICLR*.

VAE

$$\mathrm{L}(\boldsymbol{x};\theta,\varphi) = \mathbb{E}_{q_\varphi(\boldsymbol{z}|\boldsymbol{x})}[\log p_\theta(\boldsymbol{x}|\boldsymbol{z})] - KL(q_\varphi(\boldsymbol{z}|\boldsymbol{x})\,||p_\theta(\boldsymbol{z}))$$

- Maximizing L:
  - The first term encourages accurate reconstruction $\hat{\boldsymbol{x}} \approx \boldsymbol{x}$
    (đại lượng đầu muốn khôi phục chính xác mỗi mẫu x)
  - The KL term encourages $\hat{\boldsymbol{z}}$ to have a distribution similar to the prior $p_\theta(\boldsymbol{z})$
    (KL muốn phân bố của $\hat{\boldsymbol{z}}$ gần với $p_\theta(\boldsymbol{z})$)
- Training: SGD + reparameterization trick

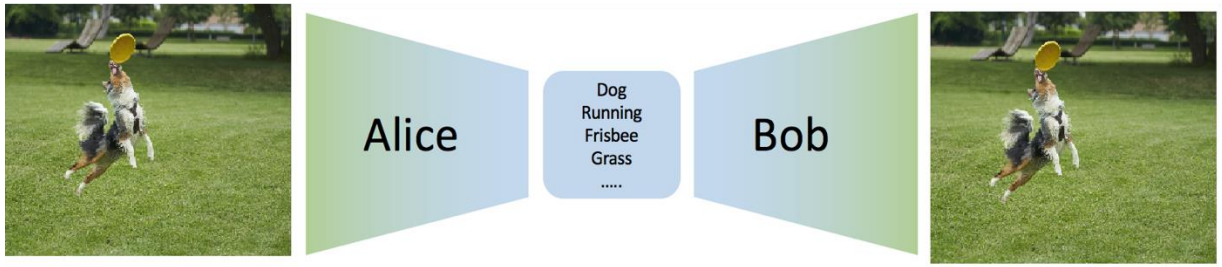


Image from Stefano Ermon

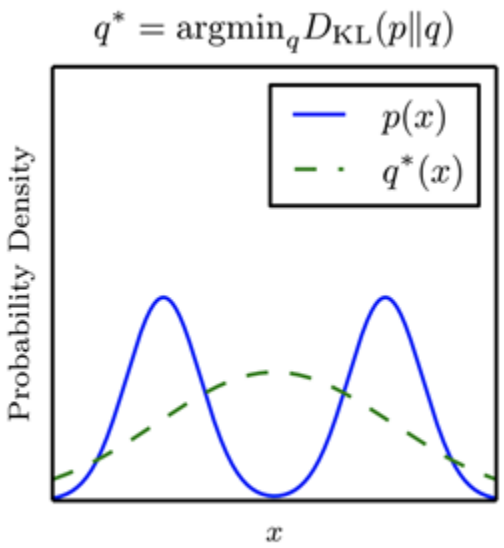
$$\boldsymbol{x} \xrightarrow{Encoder_\varphi} \hat{\boldsymbol{z}} \xrightarrow{Decoder_\theta} \hat{\boldsymbol{x}}$$

$p_\theta(\boldsymbol{x})$ $\quad$ $q_\varphi(\boldsymbol{z}|\boldsymbol{x})$ $\quad$ $p_\theta(\boldsymbol{x}|\boldsymbol{z})$

VAE: some properties

- Pros:
  - Efficient inference
  - Flexible and expressive (Universal approximator)
  - Good diversity of the synthetic samples
- Cons:
  - Blur images

VAE (2014)



Maximum likelihood

VQ-VAE (2017)